

SwiftRoute Logistics

Business Analytics Report

A complete, individually-numbered answer to all 44 business questions raised across the project's four-stage pipeline:

Excel → SQL / SQLite → Python → Power BI

Data scope: 27,979 orders · 55 drivers · 45 vehicles · 6 hubs · Jan 2023 – Dec 2024

Executive Summary

This report answers all 44 business questions raised across the SwiftRoute Logistics analytics project, each as its own individually-numbered write-up with a Finding and a Business Implication — from initial data validation in Excel, through operational SQL queries, statistical/predictive analysis in Python, to the executive Power BI dashboard. It is organized into four operational categories (Orders & Delivery, Hub Operations, Drivers & Workforce, Vehicles & Fleet) rather than by tool, so a reader can find every finding relevant to a given part of the business in one place. Numbering here follows this report's own category order and does not match the question numbers in `SwiftRoute_New_Business_Questions.docx`, which is organized by pipeline stage instead — match by question wording, not number, when cross-referencing the two.

Four findings stand out as the most consequential for SwiftRoute's operations:

- Delay is by far the strongest driver of customer satisfaction. A delayed order scores 1.56 points lower on CSAT ($p < 0.0001$), and no other factor tested — vehicle type, hub processing speed, employment type — comes close. On-time delivery is the single highest-leverage KPI to protect.
- Dallas Main Hub is structurally over capacity — running at roughly 121% of its rated monthly capacity in nearly every month across two years, while company-wide order volume has been flat (no growth trend, $R^2 = 0.008$). This is a one-time capacity/rebalancing fix, not a forecasting problem.
- Vehicle age is a strong, statistically significant predictor of breakdowns ($r = 0.65$) and does most of the predictive work on its own — the fleet's existing maintenance-alert field adds almost nothing beyond age. An age-based inspection or replacement policy is directly supported by the data.
- Delivery workload across drivers is very uneven (1,281 deliveries for the busiest driver vs. 345 for the least-used), and a real data-quality issue was found and corrected along the way: the driver name recorded on each order does not reliably match the correct driver, so every driver-level analysis in this project joins on Driver ID instead.

Roughly a third of the hypotheses tested — hub-to-hub on-time differences, vehicle type effects on delivery time, driver tenure vs. delay rate, and order-level delay prediction — did not hold up under formal statistical testing. These are reported honestly as negative findings throughout this report, because knowing where a pattern is not real is exactly as useful to the business as knowing where it is: it prevents time and budget being spent chasing noise.

Methodology & Scope

The analysis follows a four-stage pipeline, each stage using the tool best suited to the type of question being asked, in the order the work actually happens in a real analytics team:

- Excel — the four raw CSV extracts (Orders, Drivers, Hubs, Vehicles) were validated for completeness, duplicate keys, and referential integrity, and given a first analytical pass using live SUMIFS/COUNTIFS formulas. File: `SwiftRoute_Data_Preparation_QA.xlsx`.
- SQL / SQLite — the cleaned tables were loaded into a relational database and queried with joins, CTEs, and window functions (RANK, LAG) for operational questions needing more than a single aggregate. File: `SwiftRoute_SQL_Analysis.ipynb`, database: `swiftroute.db`.
- Python — pandas, scipy.stats, and scikit-learn were used to test whether patterns found upstream were statistically significant (chi-square, ANOVA, t-tests, correlation) and to build predictive/segmentation models (regression, Random Forest classification, quantile risk tiering). File: `SwiftRoute_Python_Analysis.ipynb`.
- Power BI — the validated, day-to-day operational metrics were surfaced as a live executive dashboard across four linked views (Overview, Hubs, Drivers, Vehicles), built on DAX measures. File: `LOGISTICS_DASHBOARD.pbix`.

Statistical significance throughout this report is assessed at the standard 5% threshold ($p < 0.05$). Where a test did not reach significance, this report says so explicitly rather than treating an observed difference as a real effect.

A. Orders & Delivery Performance

Q1. What data quality issues exist in the raw extracts before any analysis begins?

Answered in: Excel — DQ Checks sheet

Finding: All four tables passed core integrity checks: no duplicate Order IDs, Driver IDs, or Vehicle Codes; no orphaned foreign keys between Orders and Hubs/Vehicles; CSAT and rating fields stay within their expected 1–5 range. Cancelled orders correctly show blank delivery time/date (expected, not an error). One real issue was found and is detailed in Q32.

Business implication: Establishes that the dataset is fundamentally sound before any downstream number is trusted — a mandatory first step before building a database or dashboard on top of it.

Q2. What is total order volume for the selected period, and how does it compare month-over-month?

Answered in: Power BI — Overview (KPI card + DAX)

Finding: 1,127 orders were recorded in September 2024, up 1.8% from the previous month. Across the full two-year dataset, order volume has been essentially flat, oscillating around ~1,150 orders/month with no underlying growth trend (confirmed by linear trend analysis, $R^2 = 0.008$).

Business implication: Capacity and staffing plans should be built around a stable ~1,150 orders/month baseline rather than a growth projection.

Q3. What is the on-time delivery rate for the selected period, and how does it trend?

Answered in: Power BI — Overview (KPI card + DAX)

Finding: 79.3% of orders were on time in September 2024, down 0.6% from the previous month. The baseline delay rate across the full two-year dataset is approximately 20.4%.

Business implication: On-time rate should be treated as the top operational KPI: it is the strongest driver of CSAT found anywhere in this analysis (see Q12).

Q4. What is overall Customer Satisfaction Score (CSAT), and how does it trend?

Answered in: Power BI — Overview (KPI card + DAX)

Finding: CSAT stood at 84.1% in September 2024, down 0.4% from the previous month.

Business implication: CSAT should be monitored alongside on-time rate, not independently — the two are tightly linked (see Q12–Q13).

Q5. What is the average delivery time, and how does it trend?

Answered in: Power BI — Overview (KPI card + DAX)

Finding: Average delivery time was 35.70 hours in September 2024, down 0.5% from the previous month.

Business implication: Delivery time is a useful efficiency metric, but statistically it is not a strong lever on CSAT on its own (see Q11) — it should not be prioritized above delay reduction.

Q6. What are the top delay reasons, overall and by hub?

Answered in: Excel — Delay Reason Analysis sheet

Finding: All ten recorded delay reasons occur at a near-uniform 9–11% frequency company-wide. The Excel pivot breaks this down per hub, setting up the ranked, single-leading-cause view confirmed in Q7.

Business implication: No single delay reason dominates company-wide, so a blanket fix is unlikely to move the needle much on its own — see Q7 for the hub-level picture.

Q7. Which single delay reason is most common at each individual hub?

Answered in: SQL — window function (RANK)

Finding: The single leading delay cause differs hub to hub with no consistent pattern: two hubs are led by Road Construction, two by Package Sorting Error, and the remainder split between Vehicle Breakdown and Traffic Congestion.

Business implication: There is no single dominant delay driver to fund a company-wide fix for. A genuine root-cause investigation would need finer-grained data (e.g. minutes lost per incident, weather severity, actual construction schedules) than this extract provides before committing budget to a specific fix.

Q8. What is the order cancellation rate, and does it vary by hub or vehicle type?

Answered in: Excel — Cancellation Analysis sheet

Finding: The overall cancellation rate is 0.9% (252 of 27,979 orders). By hub it ranges narrowly from 0.6% (Fort Worth) to 1.1% (Austin); by vehicle type the spread is similarly narrow.

Business implication: Cancellations are not currently a material problem area and do not warrant a targeted intervention by hub or vehicle type on this evidence.

Q9. Does the cancellation rate cluster on particular days of the week?

Answered in: SQL

Finding: Cancellation rate by weekday ranges from roughly 0.7% to 1.1% — a narrow spread with no day standing out as a clear outlier.

Business implication: No weekday-specific staffing or process change is justified by this data; worth re-checking once cancellation reason codes are captured.

Q10. Is the hub-to-hub difference in on-time delivery rate shown on the dashboard statistically significant?

Answered in: Python — chi-square test of independence

Finding: The dashboard ranks hubs from 83.3% (Houston) down to 76.7% (Fort Worth) on-time. A chi-square test on this spread returns $p = 0.174$ — not significant at the 5% threshold.

Business implication: The ranking itself is accurate, but the gap between the best and worst hub is within the range of normal random variation given the sample sizes involved. Leadership should not treat the bottom of this ranking as proof of a genuine hub performance problem without more data.

Q11. Does vehicle type (Van/Truck/Pickup/Box Truck) have a statistically significant effect on delivery time?

Answered in: Python — one-way ANOVA

Finding: Mean delivery time is nearly identical across all four vehicle types (within hundredths of an hour of each other). ANOVA returns $p = 0.58$ — not significant.

Business implication: Vehicle-type assignment does not need to be optimized for delivery speed; other criteria (capacity, running cost, terrain) should drive that decision instead.

Q12. How much does a delay actually cost in customer satisfaction points, and is the effect statistically significant?

Answered in: Python — independent-samples t-test

Finding: A delayed order scores 2.94 on CSAT on average versus 4.50 for an on-time order — a 1.56-point gap on a 5-point scale, highly significant ($p < 0.0001$).

Business implication: This is the single largest, most reliable effect found anywhere in the dataset. Every percentage point shaved off the delay rate should show up almost directly in CSAT, making delay reduction the highest-return investment area for customer experience.

Q13. Taken together, how much of the variation in CSAT can delay, delivery time, hub processing time, and vehicle type jointly explain?

Answered in: Python — multiple linear regression

Finding: A regression on all four factors together explains 54% of CSAT's variance ($R^2 = 0.54$). The 'Is Delayed' coefficient (≈ -1.48) dominates every other variable by more than an order of magnitude.

Business implication: Confirms Q12 under proper statistical controls: if choosing between better routing to cut delays, faster hub sorting, or fleet upgrades, this analysis says delay reduction is where the CSAT return is.

Q14. Does a hub's order-processing time relate to that order's total delivery time?

Answered in: Python — Pearson correlation

Finding: Correlation is essentially zero ($r \approx 0.00$, $p = 0.68$) between hub processing time and total delivery time at the order level.

Business implication: Hub processing speed and end-to-end delivery time are functionally independent problems in this operation. Investing in faster hub sorting alone would not be expected to move the delivery-time KPI.

Q15. Can individual orders be flagged in advance as being at risk of delay?

Answered in: Python — Random Forest classifier

Finding: A classifier trained on hub, vehicle type, weekday, and driver experience/rating achieves an AUC of 0.50 — no better than random guessing.

Business implication: A genuinely useful negative result: none of the fields currently captured carry a predictive signal for delay risk. Before investing further in a delay-prediction system, SwiftRoute should capture data more likely to carry real signal — live traffic and weather conditions, precise route distance, and real-time hub queue length at dispatch time.

Q16. Does the dashboard's month-over-month order growth figure reproduce independently in raw SQL?

Answered in: SQL — window function (LAG)

Finding: Yes — a SQL query using the LAG window function recomputes the same month-over-month order figures shown on the dashboard, confirming the underlying DAX measures.

Business implication: This kind of independent cross-check is standard practice before a dashboard goes in front of leadership, and it also gives an analyst a SQL-native way to answer ad-hoc month-specific questions without opening Power BI.

B. Hub Operations

Q17. How many operational hubs does SwiftRoute have?

Answered in: Power BI — Overview / Hubs (KPI card)

Finding: SwiftRoute operates 6 hubs.

Business implication: Quick visibility into network size and coverage for planning discussions.

Q18. How do orders processed compare to each hub's rated capacity?

Answered in: Power BI — Hubs (clustered column chart)

Finding: Dallas Main Hub and Houston Hub carry the largest absolute order volumes; Dallas Main Hub's bar visibly exceeds its capacity line on the dashboard chart, the first signal of the structural issue confirmed in Q20.

Business implication: Identifies hubs operating above or below capacity, enabling better workload distribution and capacity planning at a glance.

Q19. How do hubs rank on performance (on-time %)?

Answered in: Power BI — Hubs (ranked bar chart)

Finding: Hubs rank from 83.3% (Houston) down to 76.7% (Fort Worth) on-time. As established in Q10, this spread is not statistically significant.

Business implication: Useful for routine monitoring, but not a basis for singling out a hub as underperforming without further evidence.

Q20. How many hours does each hub take to process orders, broken down by day of week?

Answered in: Power BI — Hubs (matrix chart)

Finding: Processing time by weekday is fairly stable for most hubs, with modest day-to-day variation shown in the matrix view; no hub shows a consistently slow day standing out sharply from the rest.

Business implication: Identifies any hub/day combination worth a closer operational look, though none currently stands out as a clear turnaround-time problem.

Q21. Which hubs are running over their rated capacity, based on two years of order volume?

Answered in: Excel — Hub_Capacity_Utilization sheet

Finding: Dallas Main Hub runs at approximately 121% of its rated monthly capacity on average across the full two-year total. Every other hub sits comfortably at 75–77% utilization.

Business implication: Direct input to a capacity-expansion or load-rebalancing budget conversation for Dallas Main Hub specifically.

Q22. Has hub capacity utilization been trending, or is Dallas Main Hub over capacity every month, not just occasionally?

Answered in: SQL — CTE, month-by-month trend

Finding: Dallas Main Hub is over 100% of rated capacity in nearly every month across both years of data — a structural mismatch, not an occasional spike.

Business implication: This is the clearest, most actionable capacity-planning finding in the entire project. Recommend either raising Dallas Main Hub's rated capacity (more staff/dock space) or rebalancing volume toward Houston Hub, which has the most slack in the network.

Q23. Is the Dallas Main Hub capacity problem a growth problem, or a fixed structural one?

Answered in: Python — linear trend analysis

Finding: Company-wide order volume shows no growth trend over the two-year period ($R^2 = 0.008$, slope statistically indistinguishable from flat).

Business implication: Because demand is not growing, the Dallas Main Hub capacity gap is a one-time structural fix to make, not an ongoing forecasting exercise that needs to be re-solved every quarter.

C. Drivers & Workforce

Q24. How many active drivers does SwiftRoute have?

Answered in: Power BI — Overview / Drivers (KPI card)

Finding: SwiftRoute has 55 active drivers.

Business implication: Basic workforce-planning and capacity-assessment figure used throughout the rest of this section.

Q25. How does driver experience relate to performance rating, visually?

Answered in: Power BI — Drivers (scatter plot)

Finding: The scatter plot shows an upward-sloping visual relationship between years of experience and star rating, later confirmed statistically significant in Q31.

Business implication: Gives managers a first, intuitive read on skill gaps and training needs before the formal significance test is applied.

Q26. Which drivers are contributing the most delays?

Answered in: Power BI — Drivers (bar chart)

Finding: A ranked bar chart identifies the drivers with the highest share of delayed deliveries, topped by the highest-delay driver in the dataset.

Business implication: Enables targeted coaching and performance-improvement conversations with specific individuals rather than a blanket policy.

Q27. What does an individual driver's profile look like — hire date, experience, rating, and monthly deliveries?

Answered in: Power BI — Drivers (KPI card, driver selector)

Finding: An interactive card shows hire date, years of experience, star rating, and total deliveries for the selected month, selectable by driver name.

Business implication: Gives frontline managers a consolidated view for 1:1 performance reviews and evaluation conversations.

Q28. What is the monthly trend of orders per driver?

Answered in: Power BI — Drivers (line chart)

Finding: A 12-month line chart shows each selected driver's delivery volume trend, capturing workload patterns and any seasonal demand impact at the individual level.

Business implication: Supports scheduling decisions and helps distinguish a genuine workload change from normal month-to-month noise for a given driver.

Q29. How evenly is delivery workload spread across the 55 drivers?

Answered in: *Excel — Driver Workload sheet*

Finding: Workload is highly uneven: the busiest driver (Christopher Martin) handled 1,281 deliveries over two years versus 345 for the least-used driver (mean ≈ 504).

Business implication: Uneven workload is a burnout and fairness risk, and signals inefficient dispatch logic worth reviewing.

Q30. What is the ranked distribution of driver workload, and how large is the busiest-to-least-busy gap?

Answered in: *SQL — window function (RANK)*

Finding: A ranked SQL query confirms the Excel finding with a precise spread: roughly a 3.7x gap between the busiest and least-busy driver (1,281 vs. 345 deliveries), with a standard deviation of about 214 around the mean.

Business implication: Produces an actual ranked list dispatch/HR can act on directly — either some drivers are being systematically favored for assignments, or route/hub geography is concentrating volume on a few people.

Q31. Does driver experience significantly predict performance rating?

Answered in: *Python — Pearson correlation*

Finding: Experience and performance rating show a moderate, statistically significant positive correlation ($r = 0.45$, $p < 0.001$).

Business implication: More experienced drivers genuinely tend to perform better — this supports investing in driver retention and tenure as a quality lever, not just a cost consideration.

Q32. Does the 'Driver Name' text stored on each order actually match the correct driver for that Driver ID?

Answered in: *Excel — data quality check*

Finding: No. The driver name text stored on each order does not reliably match the corresponding name in the driver master table for the same Driver ID — verified as a mismatch across all 55 drivers.

Business implication: This was the most consequential QA catch in the project. Every driver-related join used in this report, the SQL notebook, and the Python notebook uses Driver ID rather than name as a direct result, protecting the accuracy of every workforce-related finding in this section.

Q33. Does employment type (full-time / part-time / contract) relate to on-time rate or CSAT?

Answered in: *SQL — joins, aggregation*

Finding: No meaningful difference: on-time rate ranges from 78.6% (part-time) to 81.5% (contract), and CSAT is within 0.02 points across all three employment types.

Business implication: Staffing-mix decisions can be made on cost and flexibility grounds alone — there is no service-quality trade-off to weigh against them in this data.

Q34. Does driver tenure (time since hire) relate to delay rate?

Answered in: *SQL — CTE, date functions*

Finding: Delay rate is flat across tenure bands: 20.7% for drivers with 0–2 years' tenure, 21.0% for 2–4 years, and 21.5% for 4+ years.

Business implication: Tenure alone is not the lever to pull on delay rate, even though it does relate to performance rating (Q31) — worth investigating what performance rating is actually scored on, since it apparently captures something delay rate does not.

D. Vehicles & Fleet

Q35. How large is the vehicle fleet?

Answered in: *Power BI — Overview / Vehicles (KPI card)*

Finding: SwiftRoute operates 45 vehicles.

Business implication: High-level fleet-size figure for capacity and planning decisions, used as the base for every other vehicle metric in this section.

Q36. How many vehicles are currently active vs. in maintenance?

Answered in: *Power BI — Vehicles (donut chart)*

Finding: 33 vehicles are active (73.3%) and 12 are in maintenance (26.7%) at the time of the dashboard snapshot.

Business implication: A quarter of the fleet being in maintenance at any given time is a meaningful operational constraint worth tracking alongside the age-related breakdown findings in Q41.

Q37. How do total orders compare across vehicle models?

Answered in: *Power BI — Vehicles (bar chart)*

Finding: Freightliner M2 and Mercedes Sprinter handle the highest order counts among the vehicle models in the fleet.

Business implication: Identifies which models are most heavily relied upon — useful input for future fleet purchasing decisions alongside the reliability findings in Q41–Q43.

Q38. Does vehicle age relate to the number of breakdowns, visually?

Answered in: Power BI — Vehicles (scatter chart)

Finding: The scatter plot shows a clear upward visual relationship between vehicle age and breakdown count, later confirmed statistically significant in Q41.

Business implication: Gives fleet management a first, intuitive read on aging-fleet maintenance risk before the formal regression analysis is applied.

Q39. Which specific vehicles (by code) have the most breakdowns?

Answered in: Power BI — Vehicles (bar chart)

Finding: A ranked bar chart identifies the individual vehicles with the highest breakdown counts, led by FT-010 in the sample period shown.

Business implication: Supports targeted maintenance action on specific high-risk units rather than a blanket fleet-wide policy.

Q40. Which vehicle models have the most breakdowns?

Answered in: Power BI — Vehicles (bar chart)

Finding: Freightliner M2 shows the highest total breakdown count among vehicle models in the sample period shown, consistent with it also being one of the most heavily used models (Q37).

Business implication: Evaluates model-level reliability for future fleet purchasing decisions — worth checking whether this model's breakdown rate is proportional to its higher usage or a genuine reliability issue.

Q41. Does vehicle age predict breakdown risk, and does the existing maintenance-alert field add anything beyond age?

Answered in: Python — correlation + linear regression

Finding: Vehicle age is a strong, statistically significant predictor of breakdowns ($r = 0.65$, $p < 0.001$), explaining about 42% of breakdown variance on its own ($R^2 = 0.42$). Adding the existing maintenance-alert count only raises R^2 to 0.43 — it is largely redundant with age rather than an independent early-warning signal.

Business implication: Age is the best single predictor of breakdown risk SwiftRoute currently has. This supports setting an age-based proactive-maintenance or replacement threshold rather than relying on the current maintenance-alert count, which adds little predictive value beyond what age already tells you.

Q42. How are orders distributed across vehicle types (Van, Truck, Pickup, Box Truck)?

Answered in: Power BI — Vehicles (donut chart)

Finding: Vans carry the majority of order volume by type (62.7%), followed by Trucks (23.4%), with Pickups and Box Trucks making up the remainder.

Business implication: Shows fleet utilization patterns by vehicle category, useful context alongside the finding in Q11 that vehicle type does not significantly affect delivery time.

Q43. Which specific vehicles are idle or under-utilized vs. overworked?

Answered in: SQL — LEFT JOIN

Finding: 4 of the 45 vehicles (FT-014, FT-024, FT-038, FT-042) show zero deliveries against them across the entire two-year order log, while the busiest vehicle (FT-005) handled 1,263 orders over the same period (mean ≈ 616 orders/vehicle). Confirmed as a genuine utilization finding, not a data-linkage gap: every vehicle code in the order log matches the vehicle master table (100% referential match).

Business implication: The four idle vehicles are candidates to reassign, sell, or investigate (e.g. recently purchased and not yet deployed, or in extended maintenance) rather than a data problem to fix.

Q44. Can the fleet be segmented into a practical, actionable maintenance risk tier?

Answered in: Python — quantile segmentation

Finding: Splitting the fleet into age-based thirds shows the oldest third carries a disproportionate share of total breakdowns relative to the newest third.

Business implication: The 'High Risk' (oldest third) tier gives a fleet manager a ready-made inspection-scheduling priority list, and a natural input for a replacement-cycle budget conversation (e.g. replace a set number of vehicles per year from the oldest tier rather than reacting after a breakdown).

Strategic Recommendations

Based on the findings above, five actions would give SwiftRoute the highest return on the data currently available:

- Prioritize on-time delivery over every other lever tested for improving customer satisfaction — it is the only factor with a large, statistically robust effect on CSAT (Q12–Q13).
- Resolve the Dallas Main Hub capacity shortfall as a one-time structural fix (added capacity or rebalanced routing to Houston Hub), since order volume overall is flat and this is not a problem that will resolve itself with time (Q21–Q23).
- Adopt an age-based fleet maintenance and replacement policy, using the oldest-third risk tier as the starting priority list, rather than relying solely on the current maintenance-alert system (Q41, Q44).
- Review dispatch/assignment logic for drivers and reassign or investigate the 4 idle vehicles — both point to inefficiencies in how work is currently distributed across available resources (Q29–Q30, Q43).
- Before investing in an order-level delay-prediction system, capture richer data — live traffic and weather conditions, precise route distance, real-time hub queue length — since none of the fields currently available carry a usable predictive signal (Q15).

Conclusion

Taken together, the Excel QA workbook, the SQL and Python notebooks, and the Power BI dashboard form one coherent analytics project rather than four disconnected exercises. Each stage answers the kind of question it is best suited for, and each stage's output feeds credibility into the next: Excel's data-quality catch protects every later driver-related finding, SQL's operational queries surface the patterns worth testing, and Python's statistical rigor separates the real signals (delay's effect on CSAT, vehicle age's effect on breakdowns, Dallas Main Hub's capacity shortfall) from the ones that looked plausible but weren't (hub performance gaps, vehicle-type effects, tenure-vs-delay, order-level delay prediction). The dashboard is what a manager checks every morning; this report is what an analyst would use to explain, with evidence, why those numbers are what they are and where to act next.